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SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I G.CharanTeja (192372301) *(Name / Reg No) certify that this submission is my original and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary acti

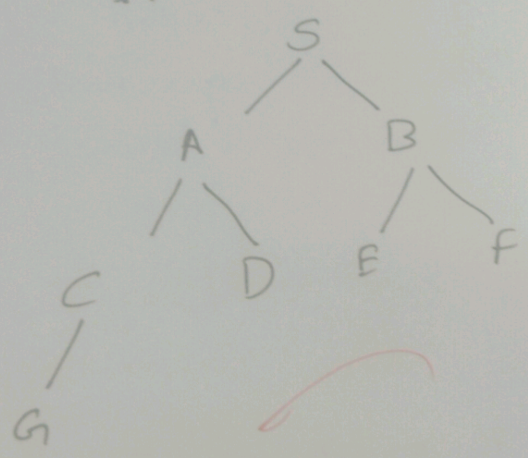
Signature of the student: G. Teja

level - 4

Given a graph with 8 nodes and a start node S and goal node G, trace depth limited search (DLS) with a depth limit of 3. show which nodes are expanded & explain. what happens if goal lies beyond DL.

Depth limited search (DLS) is an uniformed search Algorithm that is a modifed version of (DFS) in DLS, a max Depth limit is specifed before the search begins. The Algorithm explore nodes only up to this limit and does not continue be it. this helps avoid infinite pathc in very large (or) cyclic search spaces.

Graph



- Start Node: S
- goal node: G
- Depth limit: 3

DLS Algorithm

1. Start From root node (S)
2. visit one child at a time, moving as deep as possi
3. stop expanding nodes when specifed depth limit
   reached
4. if goal node is Found before reaching the
   limit, return success
5. Other wise, back track and explore the rem
   branches.

Trace of DLS

Step	Current Node	Depth	Action
1	S	0	expand
2	A	1	expand
3	C	2	expand
4	G	3	goal fo

exponded Nodes:-

S -> A -> C -> G

Conclusion:-

Depth-limited search is an efficient s
techinque when the maximum depth of
Known. It reduces unneccesary explora
may fail if the depth limit is t

the missionaries and cannibals Problem: 3 missionaries and 3 cannibals must cross a river using a boat that can carry at most 2 people, such that cannibals never outnumber missionaries either bank. Formulate the problem as a search Problem & show first three states explored.

Three missionaries and three cannibals are sta on left bank of a river. They must cross the riv using a boat that can carry at most two People. At no point should the no of cannibals exceed the number of missionaries on either bank unless there are no missionaries pre on that bank. The objective is to safely m everyone to right bank.

Problem Formulation

State Representation

each state is represented as:

(missionaries, left, cannibals, Boat pos

example:

(3, 3, L)

Initial state:

(3, 3, L)

Goal state:

(0, 0, R)

Operators (Actions):

The boat can carry

- one missionary (1M)
- Two missionaries (2M)
- one cannibal (1C)
- Two cannibals (2C)
- one missionary & one cannibal (1M,1C

Transition Model:

whenever the boat moves, the no of missonar
and cannibals on each bank is updated. A n
state is accepted only if safty condition i
satisfed

Goal Test:

The search ends succesfully when all si
Paeple reach the right bank

First three states Explored using (B

step	state	Action
1	(3,3,L)	intial state
2	(3,1,R)	Take 2 Cannibals
3	(3,2,L)	Bring 1 Canniba

Conclusion

The missionaries and Canibals Problem
lassical AI search problem that demonst
BFS systematically explores valid stat

water jug problem with a 5 gallon jug & a 3-gallon jug. How can you measure exactly 4 gallons of water using only these two jugs and a pump? List your explicit assumptions & show sequence of states from start to goa

A Person has a 5 gallon jug and one 3 ga jug. There are no measuring marks on the using only these two jugs and an unlimited water supply measure exactly 4 gallons of water

Assumptions:-

- Both jugs are initialy empty.
- unlimited water is availble
- water can be filled, emptied (or) transfer
- No measuring scale is available
- water is not lost while pouring

State Representation

each state is represented as

(Amount in 5-gallon jug, amount in 3-jug)

initial state

(0,0)

Goal state

(4,0)

Allowed Operations

- Fill any Jug completly
- Empty any Jug
- Pour water from one Jug to another u
either first Jug becomes empty or the
Jug becomes full.

Solution Steps

step	State	Action
1	(0,0)	Start
2	(5,0)	Pour into 3-gallon J
3	(2,3)	Empty 3-gallon J
4	(2,0)	Transfer remaining
5	(0,2)	Transfer remaing
6	(5,2)	Fill 5-gallon
7	(4,3)	Pour into 3 until full

Conclusion:

The water jug problem shows
sequence of valid actions can acheiv
goal using limited resources. It is
most popular examples used to
techinques in Artifical Intellig